

Institute of Actuaries of India

Subject CM2A – Financial Engineering and Loss Reserving (Paper A)

March 2022 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) III (2)
- ii) IV (2)
- iii) III (2)
- iv) II (2)
- v) IV (2)
- [10 Marks]**

Solution 2:

- i) II (2)
- ii) I (2)
- iii) I (2)
- iv) II (3)
- [9 Marks]**

Solution 3:

- i) I (2)

- ii) a) The value of the portfolio at the beginning of the month is Rs.10 Crores. The portfolio will be Rs.9 Crores or Rs.11 Crores with equal probability.

The expected utility of total wealth of the portfolio is $0.5 * (\sqrt{9 \text{ Crores}} + \sqrt{11 \text{ Crores}})$
 $= 9987.4607$
 (0.5)

The certainty equivalent C_w of the initial value and the scenario is

$$U(C_w) = 9987.4607$$

(0.5)

$$\sqrt{C_w} = 9987.4607$$

$$\Rightarrow C_w = (9987.4607)^2$$

$$\Rightarrow = 99749372$$

(0.5)

Therefore, the certainty equivalent of the scenario is

$$C_x = C_w - 10 \text{ Crores} = 99749372 - 100000000 = -250628$$

(0.5)

(2)

- b) The maximum premium the investor would like to pay to protect the portfolio is Rs.250628. (1)

- iii) If the Market is following Strong Form, then all the available information will be incorporated in the stock price and therefore, this new information will positively affect the market. The price of all stocks will go up immediately and without bias. (0.5)

If the Market is following Semi Strong Form, then all the publicly available information will be incorporated in stock price and people who are aware of this development may gain at other's cost. This can be prevented by banning Insider Trading. The fundamental analysis of the stock will provide increased value due to reduction in stock and therefore the value of the stock will go up. (0.5)

If the Market is following Weak Form, then the new information may not be reflected in the past price movements of the stock unless the market witnessed similar cut in corporate tax rate cut in the past. If any past events affected the price of the stocks and such information are reflected in the price then this new information is already incorporated in the price and therefore no change in price. (0.5)

All the above related to efficient market are true only when all investors behave rationally (0.5)

[2]

- iv) II

(2)

[9 Marks]

Solution 4:

Assuming CAPM holds, find the risk free rate using the following information

State	Probability	Risky Asset 1	Market of risky assets
1	0.2	0%	2%
2	0.25	15%	11%
3	0.05	2%	3%
4	0.5	7%	5%

Correlation between risky asset 1 and Market of risky assets = 0.9831

Solution:

Security market line

$$E1 = r + \text{Beta} * (E_m - r) \quad \text{- equation 1} \quad (0.5)$$

$$\text{Beta} = \rho * \sigma_1 / \sigma_m$$

σ_1 = std dev of risky asset 1

σ_m = std dev of the market

$$E1 = .2 * 0\% + .25 * 15\% + .05 * 2\% + .5 * 7\% = 7.35\% \quad (1)$$

$$\text{Similarly } E_m = 5.8\% \quad (1)$$

$$\sigma_1 = \sqrt{0.2 * 0\%^2 + 0.25 * 15\%^2 + 0.05 * 2\%^2 + 0.5 * 7\%^2 - E1^2} = 0.051892 \quad (2)$$

$$\text{Similarly } \sigma_m = 0.032187 \quad (1)$$

$$\text{Beta} = 1.58494 \quad (0.5)$$

Substituting all the information in equation 1

$$7.35\% = r + \text{Beta} * E_m - \text{Beta} * r \quad (0.5)$$

$$7.35\% - 9.19264\% = r * (1 - \text{Beta}) * r \quad (0.5)$$

$$r = 3.15\% \quad (1)$$

[8 Marks]

Solution 5:

- i) It is given $K=100$, $S_0=100$, $r=6\%$, $(T-t) = .25$ and $\sigma=30\%$
 Using Black-Scholes model
 Calculating $d_1 = (\ln(100/100) + (0.06 + .5 * 0.09) * 0.25) / (0.3 * \text{SQRT}(0.25))$
 $= 0.175 \quad (1)$
 $d_2 = d_1 - 0.3 * \text{SQRT}(0.25)$
 $= .025 \quad (1)$
 Value of put option
 $= 100 * e^{(-0.06 * .25)} \Phi(-0.025) - 100 \Phi(-0.175) = 5.22 \quad (1)$
- [3]**

- ii) Given that Rs. 2 is expected in 2 months the present value of dividend has to be subtracted from share price before using in the model
 The revised $S_0 = 100 - 2 * e^{-(2/12 * 0.06)}$
 $= 98.02 \quad (1)$
 Using this Share price
 Calculating $d_1 = (\ln(98.02/100) + (0.06 + .5 * 0.09) * 0.25) / (0.3 * \text{SQRT}(0.25))$
 $= 0.155$
 So $d_2 = d_1 - 0.3 * \text{SQRT}(0.25)$
 $= 0.005 \quad (1)$
 Value of put option
 $= 100 * e^{(-0.06 * .25)} \Phi(-0.005) - 98.02 * \Phi(-0.155) = 6.09 \quad (1)$

[3]
[6 Marks]

Solution 6:

- i) 3.75% --- III (1)
- ii) 0.032% --- II (2)
- iii) $500000 * (0.25 * 1.03^5 + 0.60 * 1.05^5 + 0.15 * 1.00^5) = 602,794$ --- IV (3)
- iv) Lognormal distribution has the following mean and sd

$$1.05 = \text{EXP} \left(\mu + \frac{1}{2} \sigma^2 \right) \quad \text{-- eq 1} \quad (0.5)$$

$$0.1086^2 = \text{EXP} \left(\mu + \frac{1}{2} \sigma^2 \right)^2 (\text{EXP}(\sigma^2) - 1) \quad \text{-- eq 2} \quad (0.5)$$

Substituting eq 2 to eq 1 and solving gives:

$$0.1086^2 = 1.05^2 * (\text{EXP}(\sigma^2) - 1) \quad (1)$$

$$\sigma^2 = .01064066 \text{ or } 0.10315^2 \quad (1)$$

For μ , solving from 1 equation

$$(\mu) = \ln(1.05) - \sigma^2 / 2 \quad (0.5)$$

$$\mu = 4.347\% \quad (0.5)$$

[4]

- v) Since, growth rates are independent,
A5 (which is the accumulated growth rate) follows log normal distribution ($5 * \mu, 5 * \sigma^2$)

Or

Ln(A5) (which is the accumulated growth rate) follows Normal distribution ($5 * \mu, 5 * \sigma^2$) (0.5)

Probability of a Negative remark:

$$P \{ 5 \text{ lakh} * A5 < 602794 \} \quad (1)$$

$$\Rightarrow P \{ A5 < 1.21 \}$$

$$\Rightarrow P \{ \ln(A5) < \ln(1.21) \} \quad (1)$$

$$\Rightarrow P \left\{ Z < \frac{\ln(1.21) - 5 * \mu}{\sqrt{5 * \sigma^2}} \right\} \quad (1)$$

$$\Rightarrow P \{ Z < -0.13172 \}$$

$$\Rightarrow 0.4476031 \quad (0.5)$$

If the candidate uses 601,050 from (iii) then the final answer of (v) is 0.44263981

[4]**[14 Marks]****Solution 7:**

- i) The inflation figures for the year 2018 will be ignored in the calculation as this year claims payments are fully run-off. (1)
- ii) The claim payouts given are as follows:

				Rs. In Crores
Claims paid in the year of Development				
Year of Accident	1	2	3	4
2018	275	115	62	22
2019	315	162	82	
2020	180	170		
2021	450			

The claim payouts have to be converted into 2021 inflation level. The converted payouts are as follows:

				Rs. In Crores
Claims paid in the year of Development				
Year of Accident	1	2	3	4
2018	315	126	66	22
2019	344	172	82	
2020	191	170		
2021	450			

$$\text{Where } 2018,1 = 275 * 1.05 * 1.03 * 1.06 = 315$$

$$2018,2 = 115 * 1.03 * 1.06 = 126$$

2020,1= 180*1.06

(2)

Now, the above table has to be updated for cumulative claim payouts and the updated payouts are as follows:

			Rs. In Crores	
	Claims paid in the year of Development			
Year of Accident	1	2	3	4
2018	315	441	507	529
2019	344	516	598	
2020	191	361		
2021	450			

(1)

The development factors for the payouts are as follows:

Development Factor for Year 4 = $(529/507) = 1.043$

Development Factor for Year 3 = $(507+598/441+516) = 1.544$

Development Factor for Year 2 = $(441+516+361/315+344+191) = 1.55$

(1)

Using the above development factors, the run-off triangle is completed as follows:

			Rs. In Crores	
	Claims paid in the year of Development			
Year of Accident	1	2	3	4
2018	315	441	507	529
2019	344	516	598	624
2020	191	361	417	435
2021	450	697	805	840

(1)

Now, to estimate the future inflation adjusted claims, we need to first find the estimated claims arising during each of the development years. The estimated claims arising during each of the development years are as follows:

			Rs. In Crores	
	Claims paid in the year of Development			
Year of Accident	1	2	3	4
2018				
2019				26
2020			56	18
2021		247	108	35

(1)

Now, the projected claims have to be adjusted for future inflation and the table will be adjusted as

			Rs. In Crores	
	Claims paid in the year of Development			
Year of Accident	1	2	3	4

2018				
2019				28
2020			60	21
2021		265	123	44

(1)

Therefore, the outstanding claim reserve is = $28+60+21+265+123+44 = 541$ Crores. (1)

[8]

iii) Assumptions:

- The first year is fully run-off
- Claims in each development year are a constant proportion in real terms of total claims for each accident year
- Inflation is allowed for explicitly
- Past inflation and future inflation figures are correct.

(2)

[11 Marks]**Solution 8:**

i) a) II (3)

b) II (3)

ii) I (2)

iii) IV (1)

iv) a) If entire cash is invested in asset X which gives a guaranteed return of 4% then the accumulated value at end of one year is

$$48000 \times (1.04) = 49920 \quad (1)$$

The probability of tour not happening is 1 as the accumulated value of Rs.49920 is less than Rs.50000. (1)

b) As 20% of cash is invested in asset X the accumulated value at end of 1 year is

$$20\% \times 48000 \times 1.04 = 9984 \quad (1/2)$$

The person cannot afford for tour if the accumulated value of 80% of cash invested in Y at end of one year is less than $50000 - 9984 = 40018$ (1/2)

(Let y denote the one-year return from asset Y)

$$P(80\% \times 48000 \times (1+y) < 40018) \quad (1/2)$$

$$\Rightarrow P(y < 40018 / (80\% \times 48000) - 1)$$

$$\Rightarrow P(y < 4.214\%) \quad (1/2)$$

Given $y \sim U(2\%, 6\%)$

$$P(y < 4.214\%) = \int_2^{4.214} 1/4 dy \quad (1/2)$$

$$= \frac{1}{4} [4.214 - 2]$$

$$= .55 \text{ (Rounded to 2 decimals)} \quad (1/2)$$

[5]**[14 Marks]****Solution 9:**

i) At expiry which is 1 month from now the value of the put option will be Rs.0 if share price goes up to Rs.22 and Rs.4 if share price falls to Rs.14.

To create a hedged position the value of the portfolio at expiry has to be same irrespective of share price movement. (3)

Assume 'x' software company shares are purchased.

For hedged position $22x+0 = 14x+4$

So $x = 0.5$

If 0.5 shares are purchased the value of the portfolio at expiry will be Rs.11.

The price of put option p will be $p + 20 \cdot 0.5 = 11 \cdot e^{(-0.02)}$

So $p = 0.782$

- ii) Let q denote the risk-neutral probability and with the current share price at Rs.21, q can be calculated as follows

$$22 \cdot q + 14 \cdot (1-q) = 21 \cdot e^{(0.02)} \quad (1)$$

So, $q = 0.928$

$$\begin{aligned} \text{The price of the put option is } (0 \cdot q + 4 \cdot (1-q)) \cdot e^{-0.02} & \quad (1) \\ = 0.282 & \quad (1) \end{aligned}$$

[3]

- iii) The option's delta using results from i) and ii)

$$\Delta = (0.782 - 0.282) / (20 - 21) = -0.5 \quad (1)$$

Comments:

Value of Δ shows that a delta hedged portfolio is created by purchasing 0.5 shares as calculated in i).

[2]

[8 Marks]

Solution 10:

- i) The moments of generalized pareto distributions are:

$$E(X) = k \cdot \lambda / (\alpha - 1) = (3 \cdot 75) / 3 = 75$$

$$E(X^2) = \frac{\Gamma(\alpha-2) \cdot \Gamma(k+2)}{\Gamma(\alpha) \cdot \Gamma(k)} \cdot \lambda^2 = 11250$$

Let S be the aggregate claim amount in one year. Then

$$E(S) = 120 \cdot 75 = 9000$$

$$V(S) = 120 \cdot 11250 = 1350000$$

And S follows normal distribution.

Let the loading factor be x . Then the premium collected will be $(1+x) \cdot E(S)$.

The initial capital is 1500 and the probability of ruin is 1.5%.

$$\text{Then } P(S > 1500 + (1+x) \cdot E(S)) = 0.015$$

$$\Rightarrow P\left(\frac{S - E(S)}{\text{Std Dev}(S)} > \frac{1500 + x \cdot E(S)}{\text{Std Dev}(S)}\right) = 0.015$$

$$\Rightarrow P\left(N(0,1) > \frac{1500 + x \cdot E(S)}{\text{Std Dev}(S)}\right) = 0.015$$

$$\Rightarrow \frac{1500 + x \cdot E(S)}{\text{Std Dev}(S)} = 2.17009$$

$$\Rightarrow X = 11.35\%$$

(6)

- ii) The insurer is retaining 75% of the risk. The moments of new aggregate claims paid by the insurer are as follows:

$$E(S') = 0.75 \cdot E(S) = 6750$$

$$V(S') = 0.75^2 \cdot V(S) = 759375 \quad (5)$$

Let y be the loading factor for the reinsured premium. Then the reinsurer premium will be $(1+y)*0.25*E(S)$

The condition of ruin now is

$$P(S' > 1500 + 1.15*E(S) - (1+y)*(0.25*E(S))) = 0.015$$

$$\Rightarrow P\left(\frac{S' - E(S')}{Std Dev(S')} > \frac{1500 + 1.15*E(S) - (1+y)*(0.25*E(S)) - E(S')}{Std Dev(S)}\right) = 0.015$$

$$\Rightarrow P\left(N(0,1) > \frac{1500 + 10800 - (1+y)*2250 - 6750}{Std Dev(S)}\right) = 0.015$$

$$\Rightarrow \frac{1500 + 10350 - (1+y)*2250 - 6750}{Std Dev(S)} = 2.17009$$

$$\Rightarrow y = 42.62\%$$

The insurer can pay the reinsurance premium with 42.62% loading to sustain itself in the first year with 98.5% probability.

[11 Marks]
